**DOCUMENT REVISION LIST**

Document Name : Standard Operating Procedure Template for TCS Platform Solutions

| Revision No. | Revision Date | Revision Description | Prepared By | Review By |
|---|---|---|---|---|
| 1.0 | 09th Sep 2024 | First release | | |
| 2.0 | 4th Feb 2026 | Second release – Include MS and TCS COE recommended standard steps | Gowtham Bobba<br>Satish Chodisetty | Nirupam Dandapat<br>Teng Boon Khoo |

Document Name : Hyper-V Setup build process
Document Number : v1.0

## DOCUMENT RELEASE NOTICE

This Standard Operating Procedure for KEYSIGHT TECHNOLOGIES Hyper-V Setup build process, Version 1.0, is released for use in Platform Solutions projects in Tata Consultancy Services Ltd. with effect from DD-MM-YYYY.

This document is subject to TCS Document Control Procedure.

Table of Contents


**1. Purpose** ..... 3
**2. Scope** ..... 3
**3. Prerequisites** ..... 3
**3.1 Hardware Requirements** ..... 3
**3.2 Software Requirements** ..... 3
**3.3 Licensing** ..... 3
**4. Procedures** ..... 4
**High level architecture** ..... 4
**4.1 Hyper-V Role Installation.** ..... 5
**4.2 Configure Failover Clustering** ..... 11
**4.3 Creating Logical Network in SCVMM** ..... 17
**4.4 Creating Logical Switch in SCVMM** ..... 23
**4.5 Creating VM Network in SCVMM** ..... 36
**4.6 Adding cluster/Host to SCVMM** ..... 39
**4.7 Configuring the Network adapters for new Hyper-V Host** ..... 42
**4.8 Configure ISCSI Initiator** ..... 48
**4.9 Configure Quorum Disk & Cluster Shared Volumes (CSV)** ..... 53
**4.10. Naming Network Adapters** ..... 64
**5. Post-Setup Configuration** ..... 67
**5.1 Hyper-V Setting Configuration** ..... 67
**6. Cluster Validation and recommendations** ..... 72
**6.1 Cluster Validation report** ..... 72
**6.2 NetApp Windows Host Utilities Installation Process** ..... 76
**6.3 Drivers upgrade using Dell Server Update Utility** ..... 78

## 1. Purpose

This SOP provides step-by-step instructions for building and configuring a Hyper-V cluster, deploying SCVMM for management, configuring network settings, and managing virtual machines in the new setup.
It serves as a reference for support teams to ensure consistency, reliability, and adherence to best practices during setup and configuration.

## 2. Scope

This document applies to the support of the Build & operations teams responsible for:
- Building a Hyper-V failover cluster.
- Configuring SCVMM for managing the Hyper-V environment.
- Configuring networking for the cluster and VMs.
- Managing the lifecycle of virtual machines in the newly configured environment.

## 3. Prerequisites

### 3.1 Hardware Requirements

- Minimum of two physical servers for Hyper-V cluster nodes.
- Shared storage (e.g., SAN, iSCSI).
- Redundant network interfaces for clustering, storage, and VM communication.

### 3.2 Software Requirements

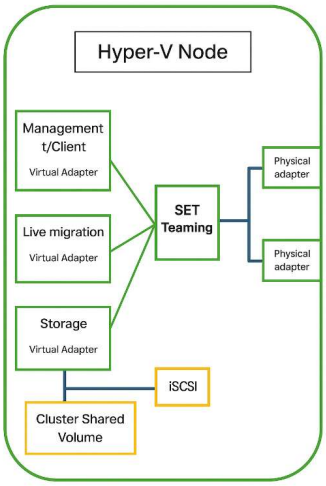
- Windows Server (supported version) with Hyper-V role installed.
- System Center Virtual Machine Manager (SCVMM) installation media or ISO.
- Active Directory and DNS integration.
- Administrative access to all servers and network devices.

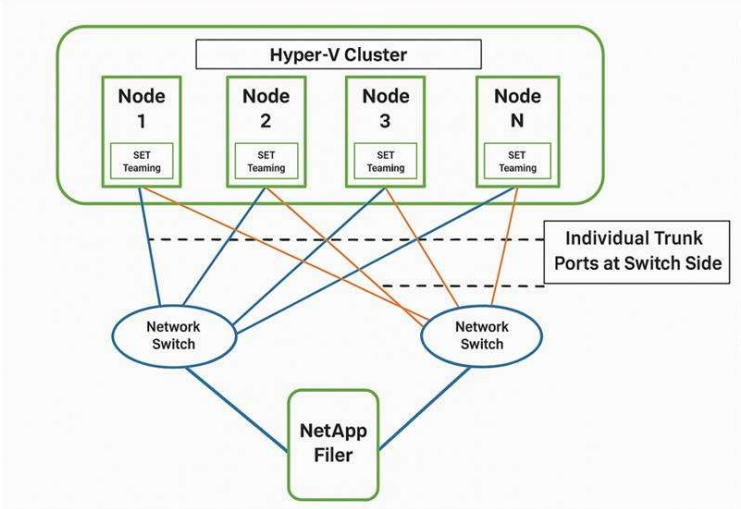
### 3.3 Licensing

- Windows Server Datacenter Edition.
- SCVMM License.

## 4. Procedures

### High level architecture

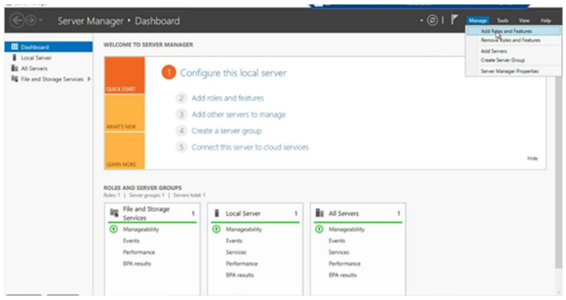


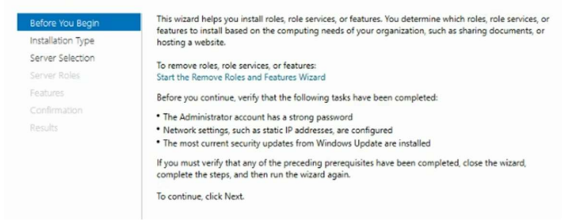



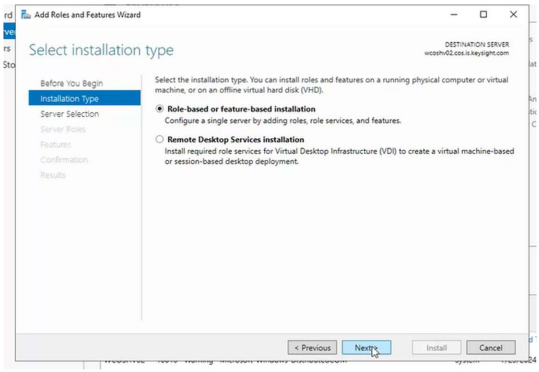
### 4.1 Hyper-V Role Installation.

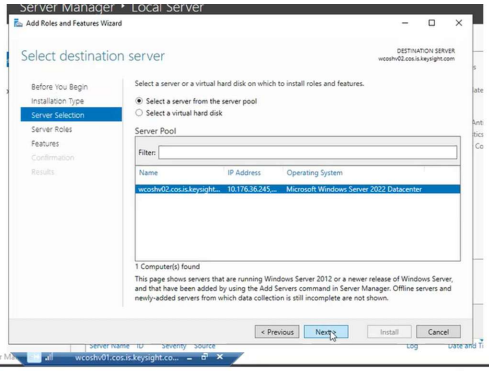
1. Log in to each physical server as an administrator.
2. Open Server Manager → Add Roles and Features.
3. Select the Hyper-V role and proceed with the installation.
4. Configure the network adapters for use with Hyper-V (i.e., to ensure that adapters are allocated for virtual switches).
5. Restart the server after installation.

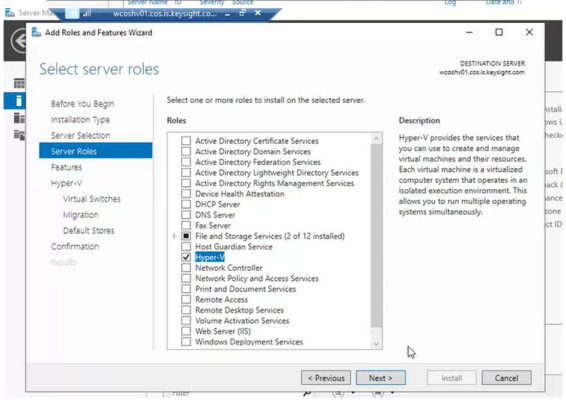
**Reference Screen Shots with Step-by-Step process:**

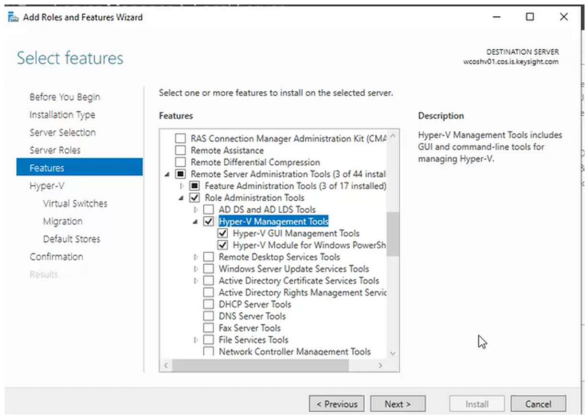


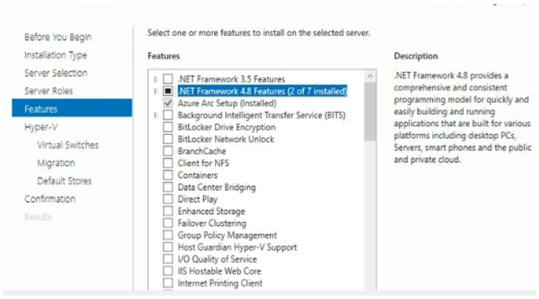



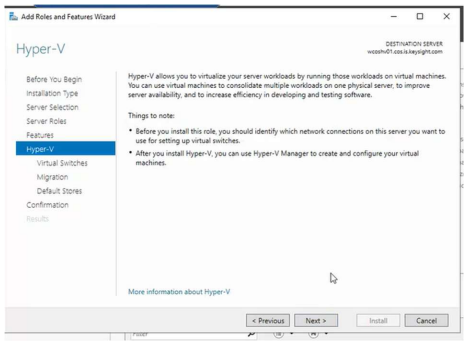

Add Roles and Features Wizard
Select installation type
DESTINATION SERVER
wcoshv02.cos.is.keysight.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).
Role-based or feature-based installation
Configure a single server by adding roles, role services, and features.
Remote Desktop Services installation
Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.
< Previous
Next >
Install
Cancel

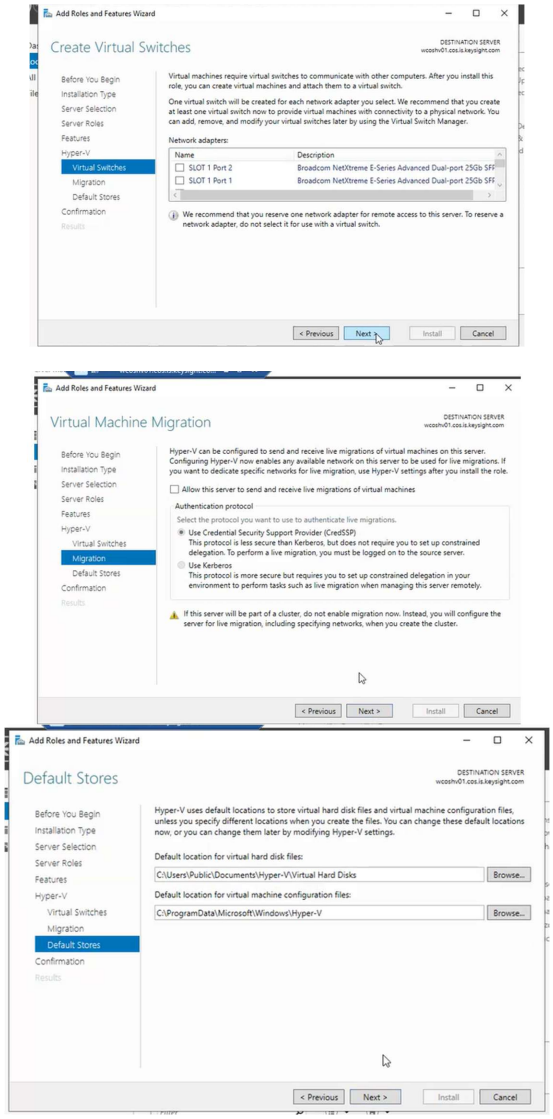

Server Manager • Local Server
Add Roles and Features Wizard
Select destination server
DESTINATION SERVER
wcoshv02.cos.is.keysight.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select a server or a virtual hard disk on which to install roles and features.
Select a server from the server pool
Select a virtual hard disk
Server Pool
Filter:
Name
IP Address
Operating System
wcoshv02.cos.is.keysight...
10.176.36.245,...
Microsoft Windows Server 2022 Datacenter
1 Computer(s) found
This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.
< Previous
Next >
Install
Cancel


Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
wcoshv01.cos.is.keysight.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
Fax Server
File and Storage Services (2 of 12 installed)
Host Guardian Service
Hyper-V
Network Controller
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Windows Deployment Services
Description
Hyper-V provides the services that you can use to create and manage virtual machines and their resources. Each virtual machine is a virtualized computer system that operates in an isolated execution environment. This allows you to run multiple operating systems simultaneously.
< Previous
Next >
Install
Cancel

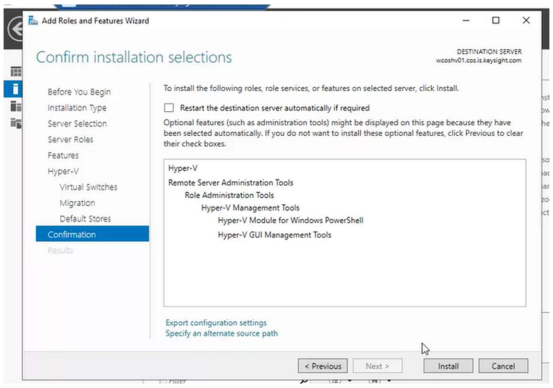

Add Roles and Features Wizard
Select features
DESTINATION SERVER
wcoshv01.cos.is.keysight.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Select one or more features to install on the selected server.
Features
Description
Hyper-V Management Tools includes GUI and command-line tools for managing Hyper-V.
RAS Connection Manager Administration Kit (CM
Remote Assistance
Remote Differential Compression
Remote Server Administration Tools (3 of 44 instal
Feature Administration Tools (3 of 17 installed)
Role Administration Tools
AD DS and AD LDS Tools
Hyper-V Management Tools
Hyper-V GUI Management Tools
Hyper-V Module for Windows PowerSh
Remote Desktop Services Tools
Windows Server Update Services Tools
Active Directory Certificate Services Tools
Active Directory Rights Management Servic
DHCP Server Tools
DNS Server Tools
Fax Server Tools
File Services Tools
Network Controller Management Tools
< Previous
Next >
Install
Cancel


Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Select one or more features to install on the selected server.
Features
Description
.NET Framework 4.8 provides a comprehensive and consistent programming model for quickly and easily building and running applications that are built for various platforms including desktop PCs, Servers, smart phones and the public and private cloud.
.NET Framework 3.5 Features
.NET Framework 4.8 Features (2 of 7 installed)
Azure Arc Setup (Installed)
Background Intelligent Transfer Service (BITS)
BitLocker Drive Encryption
BitLocker Network Unlock
BranchCache
Client for NFS
Containers
Data Center Bridging
Direct Play
Enhanced Storage
Failover Clustering
Group Policy Management
Host Guardian Hyper-V Support
I/O Quality of Service
IIS Hostable Web Core
Internet Printing Client

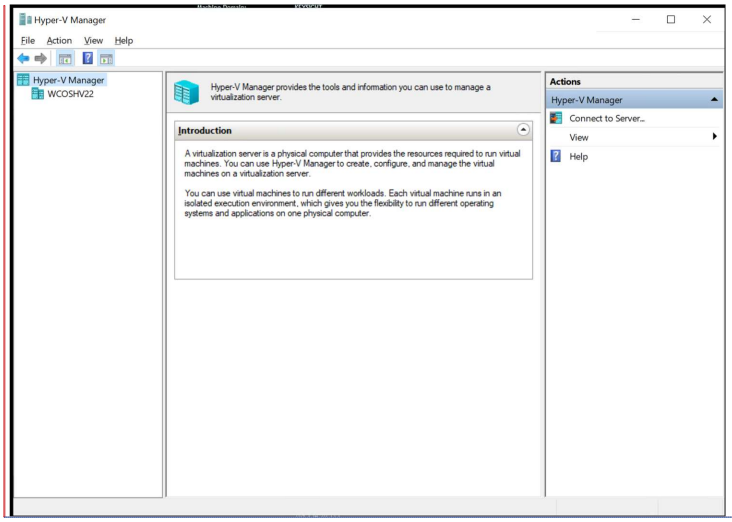

Add Roles and Features Wizard
Hyper-V
DESTINATION SERVER
wcoshv01.cos.is.keysight.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Hyper-V allows you to virtualize your server workloads by running those workloads on virtual machines. You can use virtual machines to consolidate multiple workloads on one physical server, to improve server availability, and to increase efficiency in developing and testing software.
Things to note:
Before you install this role, you should identify which network connections on this server you want to use for setting up virtual switches.
After you install Hyper-V, you can use Hyper-V Manager to create and configure your virtual machines.
More information about Hyper-V
< Previous
Next >
Install
Cancel




Path : C:\ClusterStorage\Volume (eg C:\ClusterStorage\Volume3\Productioin VMs\)



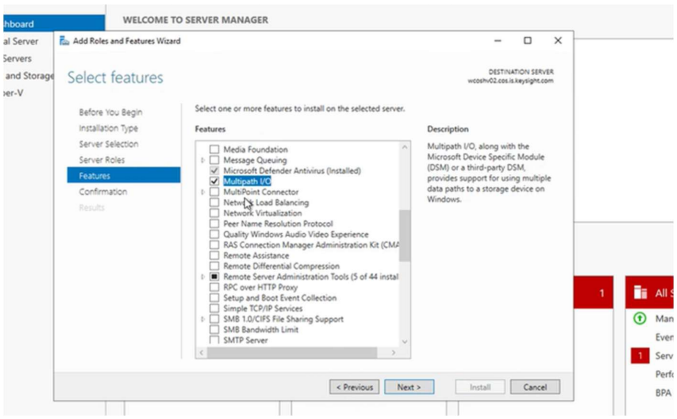

After Installed the Hyper-V role successfully then we get the Hyper-V manager console as below:

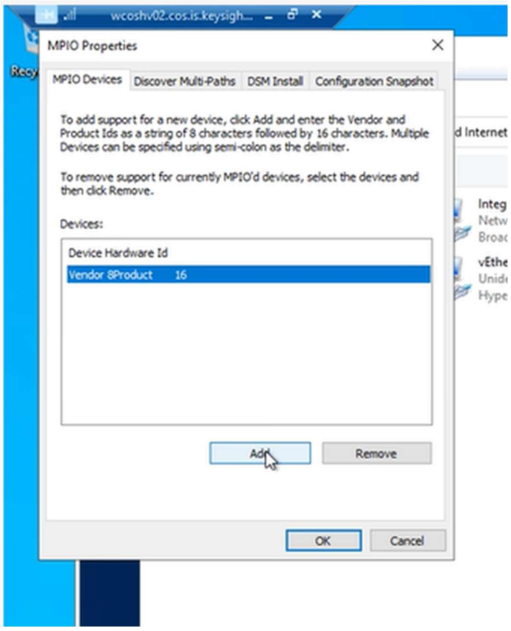


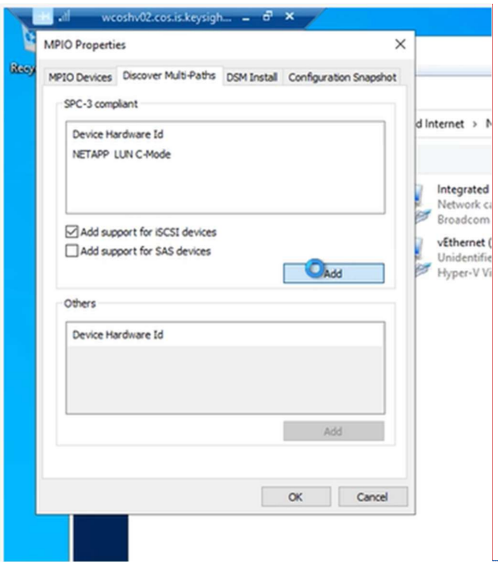

**Commented [TK1]:** Command ?diskmgmt.msc what is this for?

**Commented [GB1R2]:** Replaced screenshot with no command.

Install MPIO feature from server Manager:


WELCOME TO SERVER MANAGER
Add Roles and Features Wizard
Select features
DESTINATION SERVER
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more features to install on the selected server.
Features
Description
Multipath I/O, along with the Microsoft Device Specific Module (DSM) or a third-party DSM, provides support for using multiple data paths to a storage device on Windows.
Media Foundation
Message Queuing
Microsoft Defender Antivirus (Installed)
Multipath I/O
MultiPoint Connector
Network Load Balancing
Network Virtualization
Peer Name Resolution Protocol
Quality Windows Audio Video Experience
Remote Assistance
Remote Differential Compression
Remote Server Administration Tools (5 of 44 instal
RPC over HTTP Proxy
Setup and Boot Event Collection
Simple TCP/IP Services
SMB 1.0/CIFS File Sharing Support
SMB Bandwidth Limit
SMTP Server
< Previous
Next >
Install
Cancel


wcoshv02.cos.is.keysigh...
MPIO Properties
MPIO Devices
Discover Multi-Paths
DSM Install
Configuration Snapshot
To add support for a new device, click Add and enter the Vendor and Product Ids as a string of 8 characters followed by 16 characters. Multiple Devices can be specified using semi-colon as the delimiter.
To remove support for currently MPIO'd devices, select the devices and then click Remove.
Devices:
Device Hardware Id
Vendor 8Product 16
Add
Remove
OK
Cancel




Commented [TK2]: After add any more things to fill up?

Commented [GB2R2]: No need to fill up anything

### 4.2 Configure Failover Clustering

(This applies **only to Cluster hosts** and is **not applicable for Standalone hosts**.)

Pre-Requisites to Configure Failover Cluster

1. **Cluster Object Name and IP Address**
    a. **Reserve cluster name and IP Address in Infoblox**
        i. Use Management VLAN to reserve IP for cluster. (eg: wcoshvclu02.cos.is.keysight.com).
2. **User Privileges to Create Cluster Object in AD OU.**

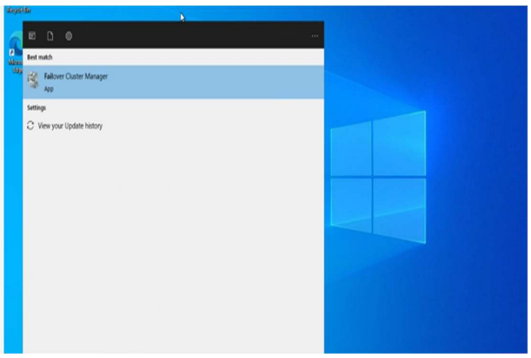
**Installation Steps:**

1. Install the Failover Clustering feature on all servers intended for the cluster:
   - Open Server Manager → Add Roles and Features.
   - Select Failover Clustering and install the feature.
2. Run the Cluster Validation Wizard:
   - Open Failover Cluster Manager.
   - Select Validate Configuration and follow the prompts.
3. Create a new cluster:
   - Open Failover Cluster Manager → Create Cluster.

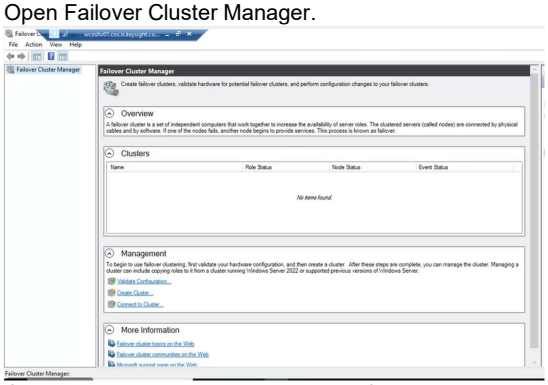
- Add all Hyper-V nodes and configure a cluster name and IP address.
- Validate that the cluster is functional.

4. Once cluster is created, please move that object into a correct OU group.

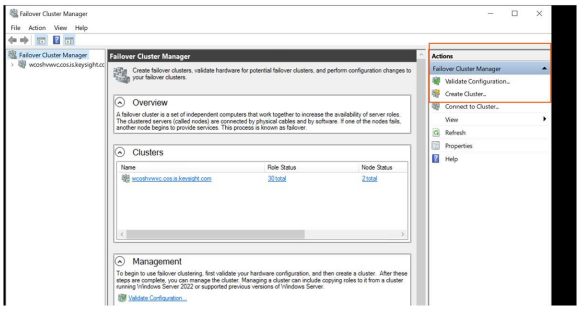
**Reference Screen Shots with Step-by-Step process:**



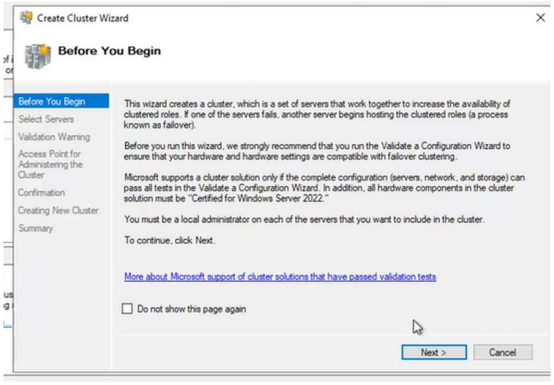

Open Failover Cluster Manager.



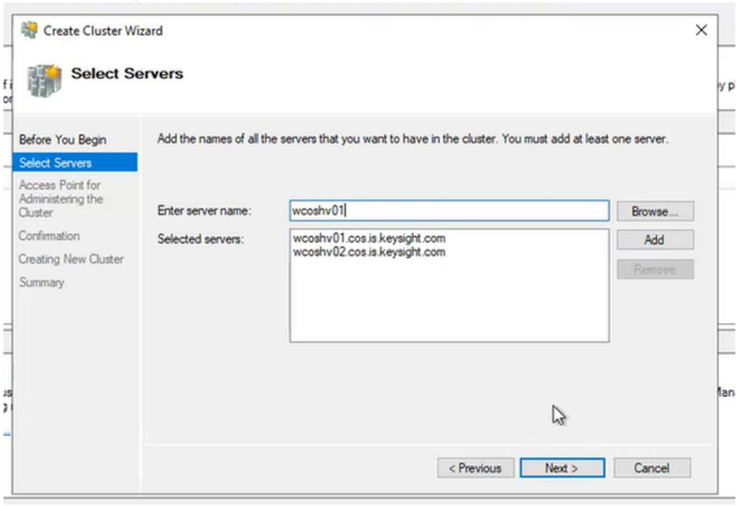

Click on create cluster under Actions from right-side menu panel.



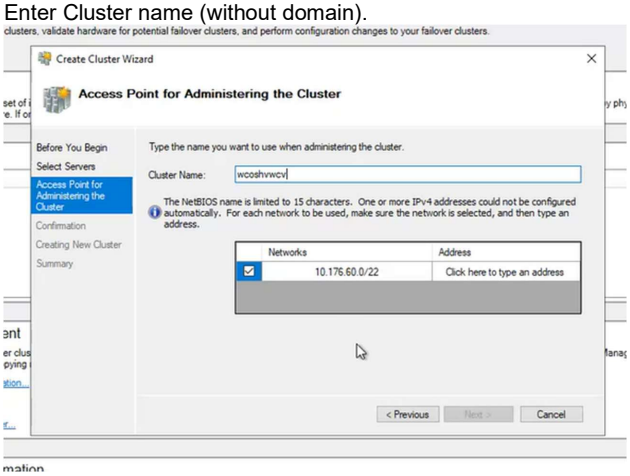

Follow the creation of cluster wizard.



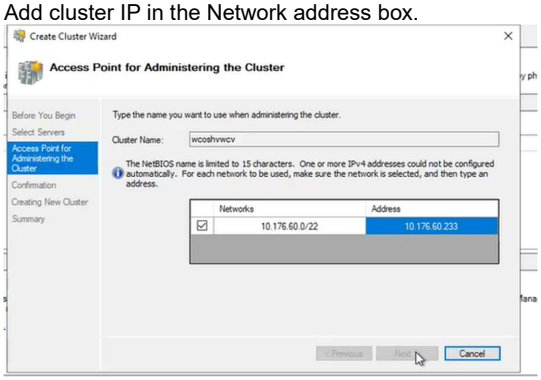

Add hosts FQDN and click next



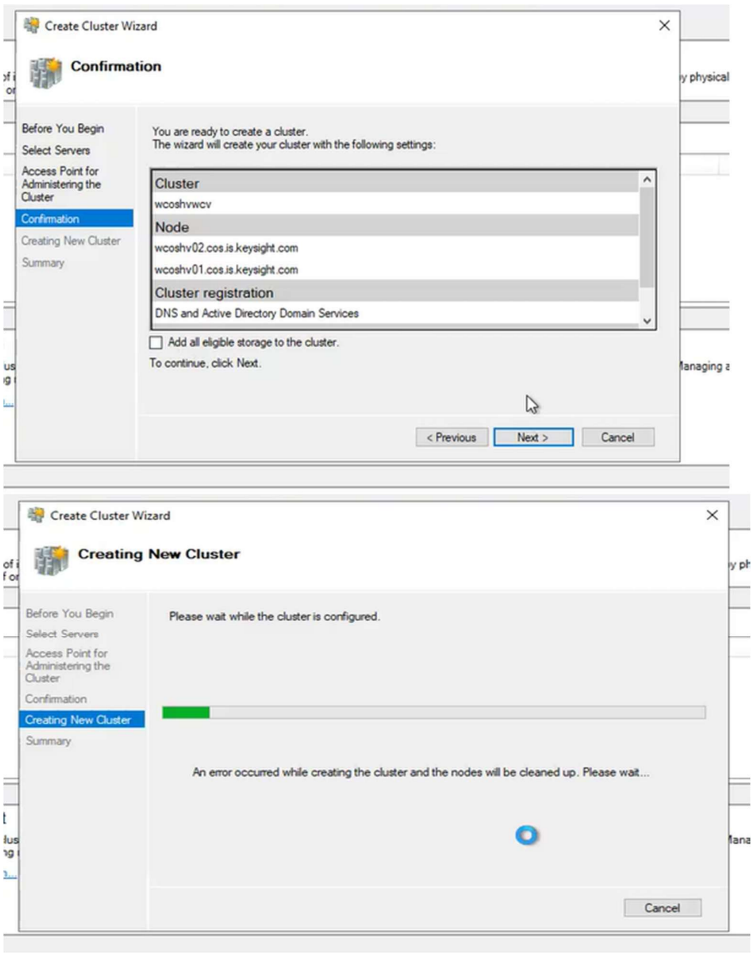
Enter Cluster name (without domain).



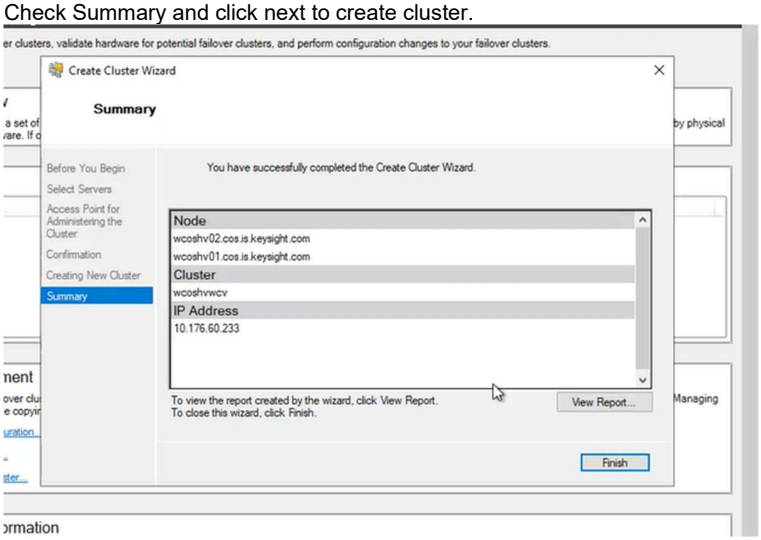

Add cluster IP in the Network address box.



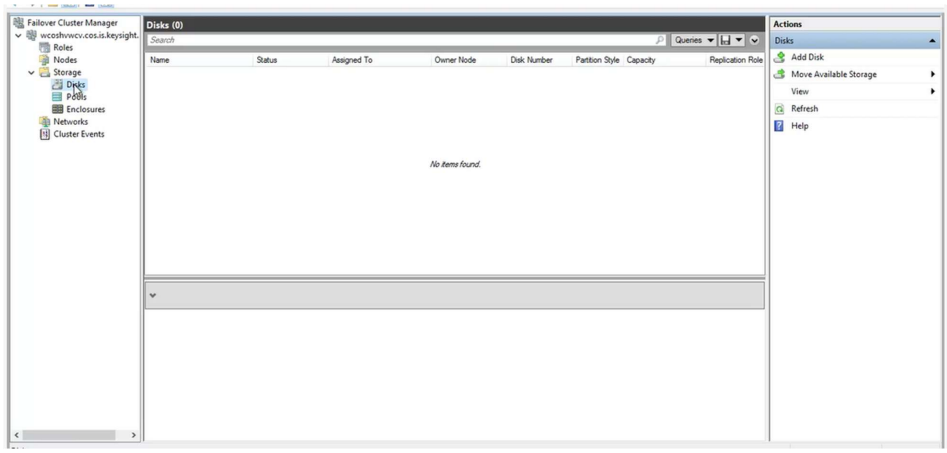

Click next to create a cluster.


Create Cluster Wizard
Confirmation
Before You Begin
Select Servers
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
You are ready to create a cluster.
The wizard will create your cluster with the following settings:
Cluster
wcoshvwcv
Node
wcoshv02.cos.is.keysight.com
wcoshv01.cos.is.keysight.com
Cluster registration
DNS and Active Directory Domain Services
Add all eligible storage to the cluster.
To continue, click Next.
< Previous
Next >
Cancel
Create Cluster Wizard
Creating New Cluster
Before You Begin
Select Servers
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
Please wait while the cluster is configured.
An error occurred while creating the cluster and the nodes will be cleaned up. Please wait...
Cancel

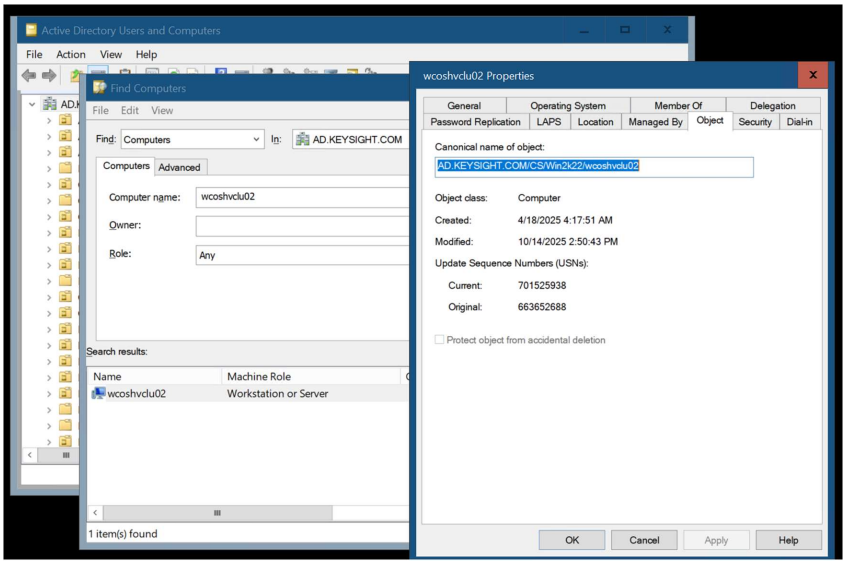
Check Summary and click next to create cluster.




Once Cluster created you can find cluster name under Failover cluster Manager from left side menu.

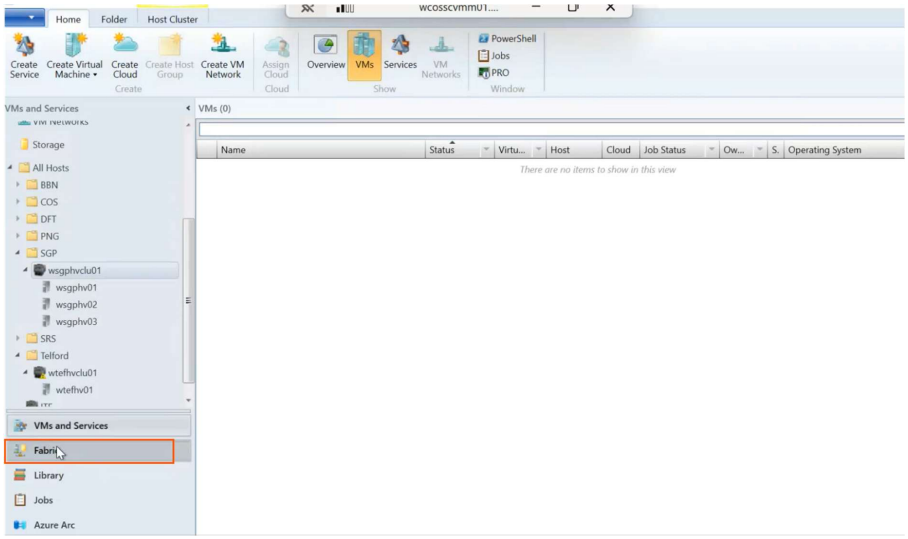



Move the created cluster object to the appropriate OU group.

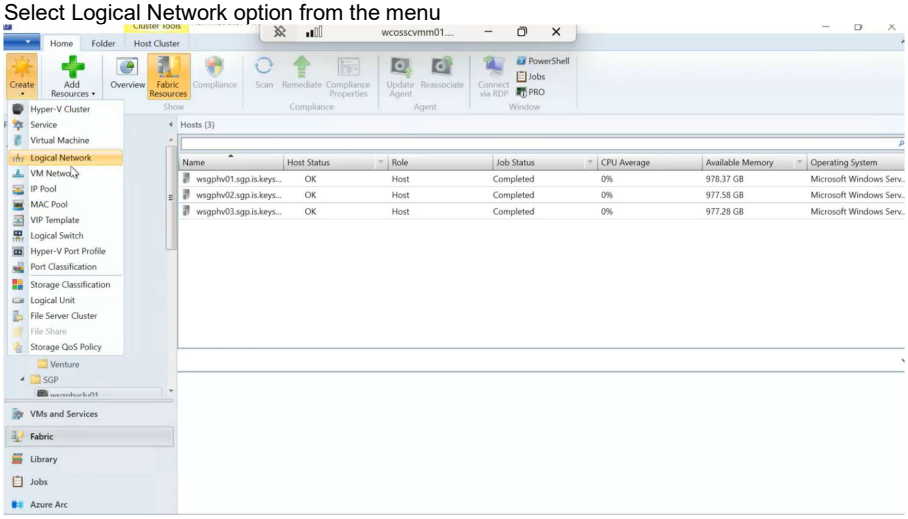



### 4.3 Creating Logical Network in SCVMM

Navigate to Fabric from the left side menu panel.

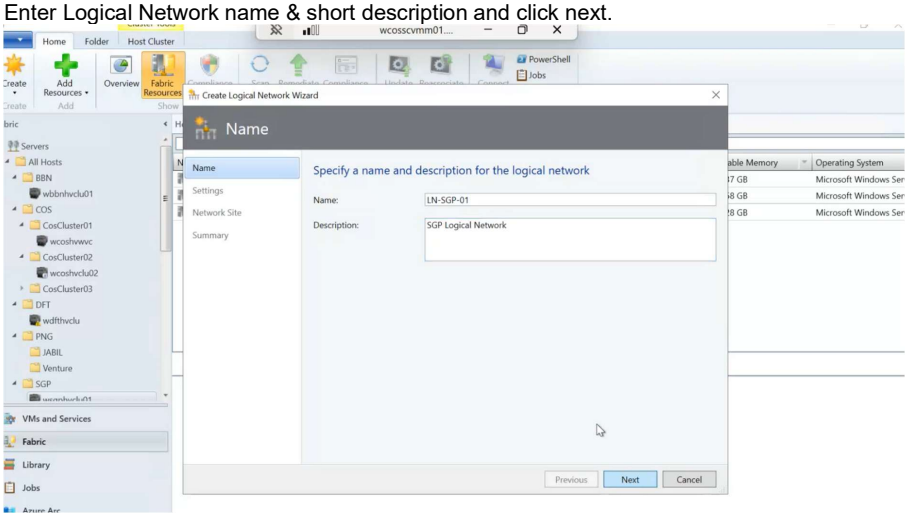



Select Logical Network option from the menu

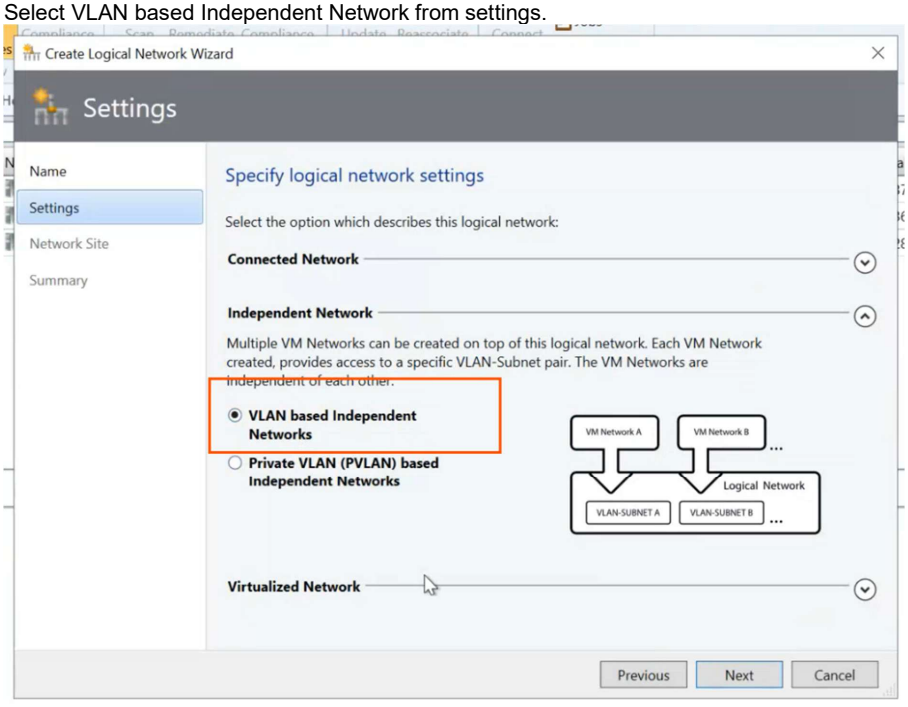



Follow create Logical Network Wizard.

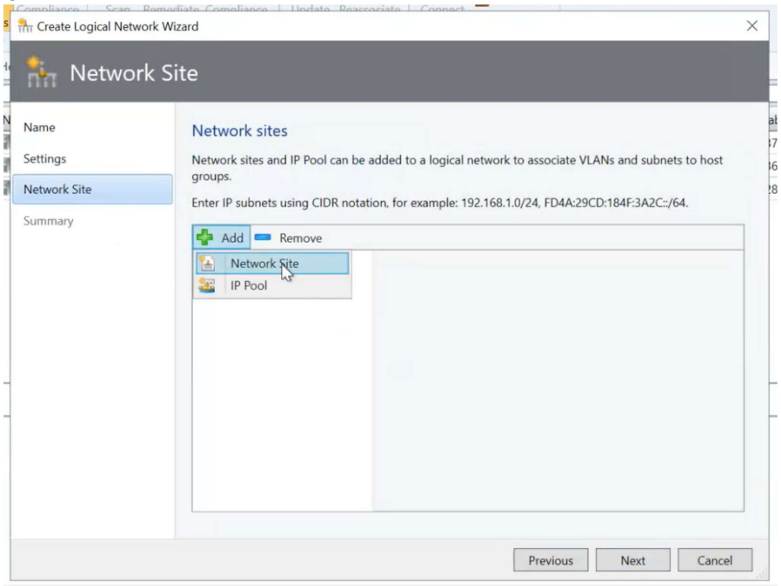
Enter Logical Network name & short description and click next.




Select VLAN based Independent Network from settings.



Click on Add new and select Network Site and enter network site name.




Select Host group

